Quiz 4

Q1. [6 Marks] An e-commerce database tracks product inventory. Two transactions run concurrently:

Transaction T1 (Warehouse Manager): Updates the stock count for Product A.
Transaction T2 (Sales Analyst): Checks the stock count for Product A twice to confirm availability.

- T1 reads Product A's stock as 50.
- T2 reads Product A's stock as 50.
- T1 updates Product A's stock to 40 and commits.
- T2 reads Product A's stock again and now sees 40.
- T2 updates Product A's stock to 45 (**overwriting T1's change**).

Questions:

1. [2 Marks] Identify the concurrency problems in the scenario. Provide your reason (at max 1 sentence).
2. [2 Marks] State which ACID property is being violated for each problem.
3. [2 Marks] How can these issues be prevented?

1. Non Repeatable Read: T2 reads twice and sees different values.

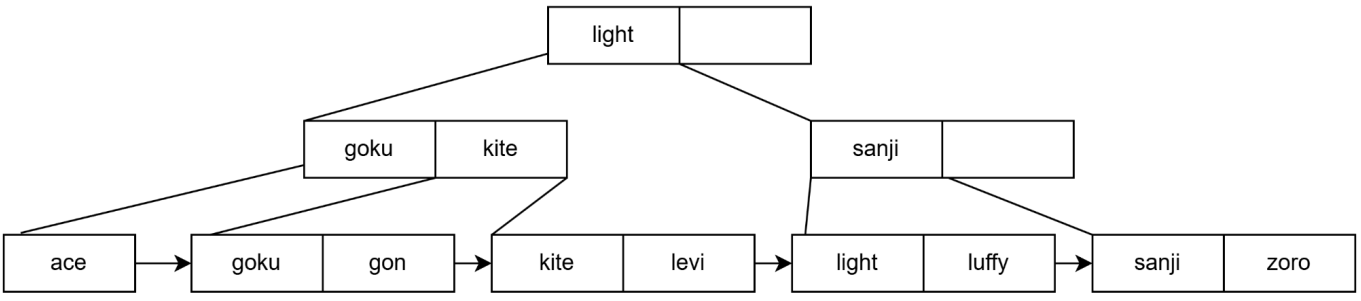
   Lost Update: T2 overwrites T1's value.

2. Isolation and Consistency
3. Setting Isolation level to Serializable

Q2. [4 Marks] For the following B+ tree of order n = 4, simulate the <u>deletion</u> of these keys in this order:

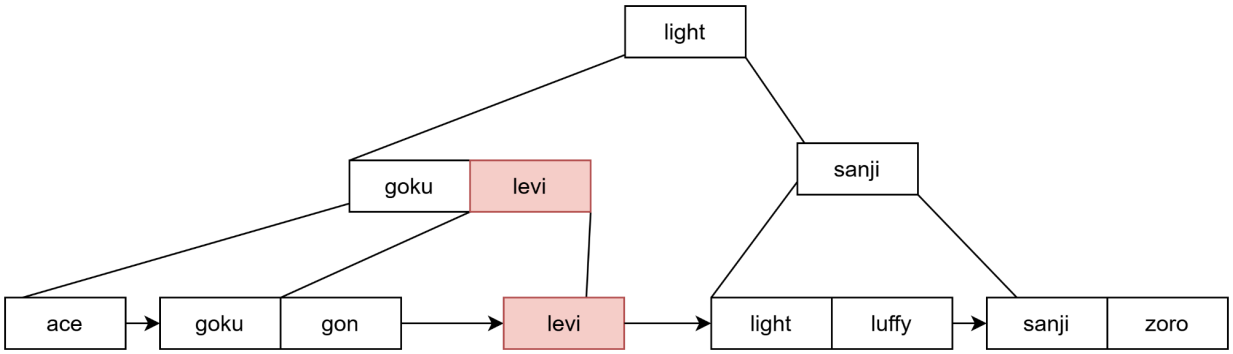
a) Delete kite
b) Delete levi
c) Delete gon
d) Delete goku

Note: All operations are to be performed one after the other on the same tree.

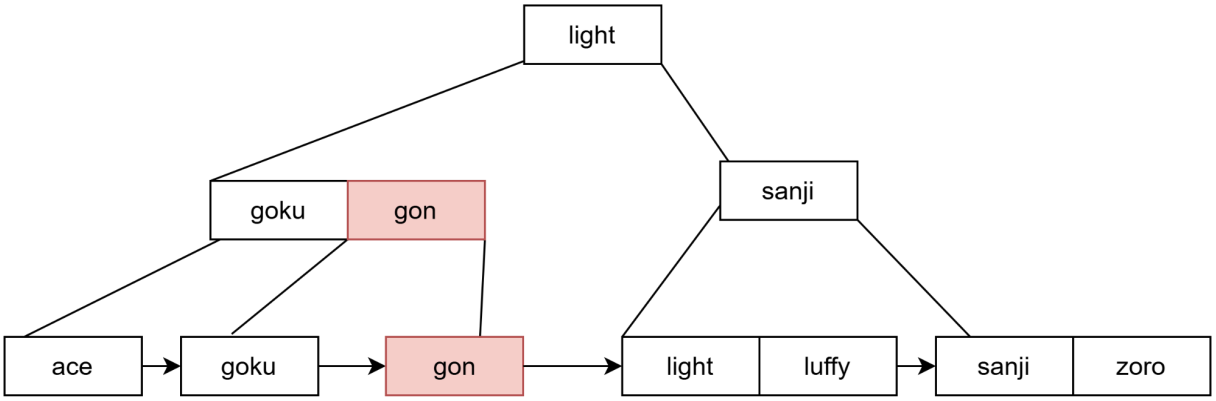


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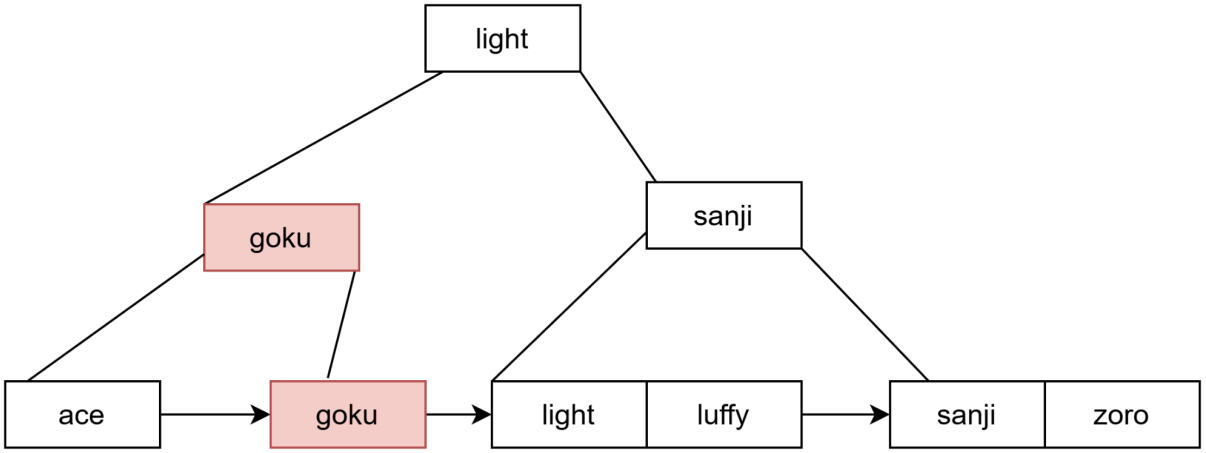
a) Delete kite



b) Delete levi



c) Delete gon



d) Delete goku

